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Original Research Paper

Exploring injury severity of children and adolescents involved in traffic crashes in Greece

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HIGHLIGHTS

- Injury severity of children and adolescents involved in crashes are explored.
- To account for potential unobserved heterogeneity, mixed logit models are applied.
- Crashes outside urban areas increase severity of children and adolescents.
- Night crashes and crashes involving two-wheelers also lead to more severe injuries.
- The likelihood ratio test confirmed the temporal instability of risk factors.

ARTICLE INFO

Article history:

Received 3 July 2019

Received in revised form

14 July 2020

Accepted 15 July 2020

Available online 28 December 2020

Keywords:

Traffic engineering

Traffic crash

Injury severity

Children

Adolescents

Unobserved heterogeneity

ABSTRACT

For the purpose of exploring the factors affecting injury severity of children and adolescents involved in traffic crashes in Greece, disaggregate crash data including 13,431 involving children and adolescents from all regions of Greece for the period 2006–2015 were utilized. In order to identify factors affecting injury severity and account for potential unobserved heterogeneity, a series of mixed logit models were utilized. To explore and address potential temporal instability of crash-related risk factors, the likelihood ratio test was applied. Results indicated that night crashes, crashes outside urban areas as well as crashes involving bicycles or powered-two-wheelers are associated with higher injury severity of children and adolescents. Interestingly, crashes involving pedestrians are associated with lower injury severity than head-on collisions and run-off-road collisions with fixed objects. Side and sideswipe crashes also result in lower injury severities. The likelihood ratio test indicated that crash-related factors are instable when comparing the models utilizing data before and after 2010 respectively. This study contributes to the current knowledge in the field, as to the best of our knowledge this is the first study that addresses unobserved heterogeneity when analyzing child and adolescent injury severity. Overall, the findings of this study provide useful insights and could assist in unveiling crash risk factors and prioritize programs and measures to promote road safety of children and adolescents.

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Peer review under responsibility of Periodical Offices of Chang'an University.<https://doi.org/10.1016/j.jtte.2020.07.005>2095-7564/© 2021 Periodical Offices of Chang'an University. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

1. Introduction and background

Crash injury severity analysis has received a lot of attention from researchers worldwide, with many types of studies across different countries and sample frames. International literature has identified various risk factors associated with crash injury severity level sustained by road users, such as individual characteristics (age, gender, experience, alcohol), speeding, time (day or night), weather conditions, area type, etc. Great efforts have been devoted to the analysis of vulnerable road users' injury severity, such as pedestrians, cyclists and motorcyclists (Lin et al., 2019; Moore et al., 2011; Pour-Rouholamin and Zhou, 2016; Theofilatos and Ziakopoulos, 2018; Xie et al., 2018). For instance, Zhai et al. (2019) analyzed pedestrian injury severity with real-time weather data and found that high temperatures and increased rainfall were associated with a higher probability of fatal and severe crashes. Shaheed and Gkritza (2014) explored single-vehicle motorcycle crash severity indicating several specific factors such as speeding, run-off-road, riding without a helmet, etc.

However, one aspect that is often neglected is the injury severity analysis of children and adolescents involved in road crashes. This is considered an important issue worth to investigate due to the high traffic mortality of children worldwide. According to WHO (2008), road traffic injuries are the leading cause of death in 10–19 years old globally. In this report, it is also suggested that even in European Union countries, where rates are not as high, road traffic injuries still account for 20% of childhood injury deaths.

A few studies have had a focus on child pedestrian injury analysis (Adesunkanmi et al., 2000; Bennet and Yiannakoulis, 2015; Koopmans et al., 2015). For instance, Bennet and Yiannakoulis (2015) explored motor-vehicle collisions involving child pedestrians walking to school in Hamilton, Ontario, Canada and suggested that child pedestrian injuries at intersections are correlated with the type of intersection control, traffic volume and land use. Another study by Peek-Asa et al. (2010) suggested that rural crashes involving teens were nearly five times more likely to result to a fatal/severe crash than urban teen crashes. Koopmans et al. (2015) analyzed age-based urban pedestrian crash characteristics and compared them with car crashes in order to identify crash characteristics associated with injury severity. In that study, overall incidence was higher for child versus adult pedestrians, but the fatality rate for children was lower. In addition, daylight crashes were associated with less severe injuries of child pedestrians than night crashes.

Research on bicycle injury severity of children and adolescents is relatively limited (Hagel et al., 2015) and it is

suggested that in bicycle collisions with motor vehicles, male cyclists, as well as those not wearing a helmet, were more likely to sustain increased injury severity. Collisions when cycling on a paved surface or cycling to school or work tend to decrease injury risk. On the other hand, more research has been carried out focusing on injury severity when children and adolescents are vehicle occupants (García-España and Durbin, 2008; Koekemoer et al., 2017; Olsen et al., 2010; Skjerven-Martinsen et al., 2014; Zaveri et al., 2009). For example, García-España and Durbin (2008) found that injury risk of car occupants is higher than sport utility vehicles (SUVs), large vans and minivans (but not pickup trucks). Peek-Asa et al. (2010) suggested that fatality rates of teen drivers were generally higher for rural than urban settings, for all causes.

There are several studies that have linked child occupant injury severity with restraint use or misuse and seating position of children (Arbogast et al., 2005; Brown et al., 2006, 2010; Decina and Lococo, 2005; Durbin et al., 2005; Viano and Parenteau, 2008). Other common risk factors include driver restraint use, light conditions and age as well as pedestrian collisions with bicycles. For example, Skjerven-Martinsen et al. (2014) showed that among multi-vehicle crashes occurring during daylight, 12% of child occupants sustained abbreviated injury scale (AIS) 2+ injuries (moderate to severe injuries with at least one body section defined as injured), whilst the respective percentage while traveling in darkness is 38%. However, in the same study, authors did not observe any significant influence of gender.

According to García-España and Durbin (2008), head injuries were the most common types of injury (60%), while 21% of 8–12 years old children did not use the shoulder portion of the car seat belt nor did incorrectly place it (behind their back or under their arm). A general remark is that obesity was found to be highly correlated with pediatric injury severity and distribution of injuries or restraint use (Zaveri et al., 2009). A recent study by Duddu et al. (2019) investigated the risk factors associated with injury severity of teen drivers by applying a partial proportionality odds model. The authors found that sports utility vehicles and pickup trucks are more likely to be associated with severely injured teens (compared to passenger cars). Furthermore, teen drivers are more likely to be severely injured on weekdays and peak hours.

Overall, it is observed that there is a limited number of studies focusing on analyzing the impact of various risk factors on child and adolescent injury severity. It is also noteworthy that low- and middle-income countries are underrepresented, although these countries account for 93% of child road traffic deaths (WHO, 2008). In terms of data analysis methods, it is noteworthy that although most of relevant research in the field of child and adolescent injury risk is robust, it mostly

explores the topic by applying relatively simple statistical methods, such as chi-square tests, relative risk ratios or other descriptive analyses. On the other hand, the common approach of those studies which utilized more advanced statistical modelling was the application of fixed effects models (e.g., fixed effect binary logistic model).

A very important issue to consider when analyzing crash injury severities is the potential unobserved heterogeneity. It is noted that [Mannering et al. \(2016\)](#) emphasized the strong need to account for unobserved heterogeneity when analyzing crash injury severity data. It is observed however, that so far the issue of unobserved heterogeneity has not been accounted for when investigating children and adolescent injury severity outcomes. However, this is not the case when aiming to explore crash severities in general, as a great number of studies attempted to address the issue of potential unobserved heterogeneity ([Cerwick et al., 2014](#); [Fountas and Anastasopoulos, 2017](#); [Fountas et al., 2018a, b](#); [Kim et al., 2010, 2017](#); [Shaheed and Kkritza, 2014](#); [Xu et al., 2016](#)).

To the best of our knowledge, another important issue that has never been examined when exploring children and adolescent injury severity is the potential temporal instability ([Al-Bdairi et al., 2020](#); [Behnood and Mannering, 2015, 2019](#); [Mannering, 2018](#)), i.e., the potential temporally unstable behavior of crash-related factors. This topic has recently gained much attention from researchers and the importance of temporal instability is stressed by [Mannering \(2018\)](#), who stated that “the possibility of temporal instability has profound implications for current safety practice and allocation of funds for safety improvements”.

Consequently, our study adds to current knowledge by: i) exploring child and adolescent injury severity of road crashes in Greece for a 10-year period (2006–2015), and ii) addressing potential unobserved heterogeneity. Two possible injury outcomes were considered, namely killed/severely injury and slight injury. It is widely common for crash injury severity studies to include two possible outcomes ([Al-Ghamdi, 2002](#); [Sze and Wong, 2007](#); [Theofilatos et al., 2012](#); [Yu and Abdel-Aty, 2014a, b](#)).

To the best of knowledge, this is the first study to address the issue of potential unobserved heterogeneity when exploring child and adolescent injury severity. More specifically, in order to account for potential unobserved heterogeneity, a mixed logit model is applied, in which the constant term follows a normal distribution across all Greek counties. Furthermore, a “year” indicator variable (before or after 2010) was included as a candidate independent variable, so as to consider the potential impact of the financial crisis in Greece that started in late 2010/early 2011 and ended in late 2015/early 2016. Potential temporal instability of crash injury related factors is also investigated, hence separate models for before and after 2010 were also applied and compared with the full holistic model (2006–2015) by applying likelihood ratio test.

2. Materials and methods

2.1. Data collection and processing

The required crash data were collected from the Greek crash database SANTRA, which is provided by the Department of

Transportation Planning and Engineering of the National Technical University of Athens. The SANTRA database consists of all crash data in Greece, which are filled in high detail by the police immediately after the occurrence of a crash. It is noted that disaggregate data require specific treatment because they are confidential and must be used only for a scientific and research purposes.

In our analysis, the crash dataset of interest involved all injured persons under 18 years old (children and adolescents) who were involved in traffic crashes occurred in all Greek regions (counties) for a 10-year period, namely 2006–2015. A total of 13,431 children and adolescents were involved in total. The response variable is injury severity, which is a binary variable in nature. This study aims to simultaneously analyses the injury severities of all persons under 18 years old involved in a crash, consequently each row in the database represents an injured person. Two injury severity levels were considered: 0 = slight injury (SI) and 1 = killed/severe injury (KSI). It is noted that the Greek crash database SANTRA does not include information about property damage only (PDO) crashes.

The aim is to investigate factors affecting the outcome of a child/adolescent injury (KSI versus SI), given that the crash has occurred. Hence, we followed the traditional approach to investigate injury severity; that is a comparison between KSI and SI cases. The comparison with adult injuries was out of the scope of this paper, therefore no injury data for adults were considered. Additionally, since this is the first study that attempts to provide insights on this issue in Greece, it was not aimed to compare children with adolescents.

It is noted that in the Greek crash database, fatalities are defined as all persons killed within 30 d after the crash, whilst severe injuries are all persons hospitalized for at least 24 h. Lastly, slight injuries are all other persons reported by the police as injured without hospitalization. Other individual and crash specific variables that were considered include gender, weather, area type, location, vehicle type, illumination and crash type. The year of the crash was also considered in order to explore the temporal evolution of injuries. More specifically, the time periods before and after the financial crisis that heavily occurred after 2010 in Greece were examined.

2.2. Methodology

To account for unobserved heterogeneity, a random parameter binary logistic model (mixed logit) was applied. In random parameter models, parameters may vary across observations, in a sense that they follow a distribution, e.g., normal, uniform, etc. Following [Washington et al. \(2011\)](#), the random-parameter model has for observation n outcome probabilities defined as

$$P_n^m(i) = \int_x P_n(i)f(\beta|\varphi)d\beta \quad (1)$$

More specifically, for a mixed model (a model with a mixing distribution), $P_n^m(i)$ is the probability of observation n having a discrete outcome i , I is all possible outcomes for observation n , X_{in} is a vector of observable characteristics (covariates) that determine discrete outcomes i for observation n , β is a vector

of estimable parameters for discrete outcome i , $f(\beta|\varphi)$ is the density function of β , and φ is a vector of parameters of the density function (e.g., mean and variance). Consequently, Eq. (1) can be rewritten as

$$P_n^m(i) = \frac{\int \frac{\exp(\beta_i X_{in})}{\sum_I \exp(\beta_i X_{in})} f(\beta|\varphi) d\beta}{\sum_I \exp(\beta_i X_{in})} \quad (2)$$

In our approach, a random parameters binary logistic model (mixed logit model) in which the constant term is following a normal distribution is applied and its goodness-of-fit is compared with the fixed effects model. The goodness-of-fit of the models can be assessed with the likelihood test ratios of the full model (L_f) and the empty model (L_0). Further model comparisons will be carried out by using the Akaike information criterion (AIC) and Bayesian information criterion (BIC).

In order to examine potential temporal instability across different time periods (before and after 2010), and more specifically to justify estimating separate models for two time periods instead of an overall one, the likelihood ratio test is performed, following Washington et al. (2011) and Al-Bdairi et al. (2020).

$$\chi^2 = -2[\text{LL}(\text{overall_model}) - \text{LL}(\text{before_2010}) - \text{LL}(\text{after_2010})] \quad (3)$$

where $\text{LL}(\text{overall_model})$ is the log-likelihood at convergence of the overall model (2006–2015), $\text{LL}(\text{before_2010})$ is the log-likelihood at convergence of the model regarding 2006–2010 and $\text{LL}(\text{after_2010})$ is the log-likelihood at convergence of the model regarding 2011–2015. As suggested by Al-Bdairi et al. (2020), the same variables were used in all three models.

3. Results and discussions

A preliminary analysis was firstly carried out. Overall, the total number of injured children and adolescents involved in road crashes in the 51 counties of Greece was 13,431, out of which 11,702 were classified as slightly injured (SI) and 1729 as killed/severely injured (KSI). Table 1 provides an illustration of the variables considered, potential values as well as indicative descriptive statistics. Table 2 illustrates the temporal evolution of crash severity by year, showing a steadily decreasing number of both slight injuries and severe injuries/fatalities.

Afterwards, the mixed logit model was applied in order to analyze the impacts of crash and individual specific variables on injury severity of children and adolescents. 200 Halton draws were used, whilst the constant term was set as random following a normal distribution. The normal distribution was found to be the best as this was the model with the best goodness-of-fit. In addition, a number of studies in the field stated that the normal distribution is the most appropriate for the random parameters (Behnood and Mannering, 2017; Moore et al., 2011). The summary results of the holistic model for the period 2006–2015, as well as for the separate models for 2006–2010 and 2011–2015 periods are presented in Tables 3–5.

Table 1 – Variables considered, potential values and descriptive statistics.

Variable	Number of cases
Severity_slightly injured	11,702
Severity_killed/severely injured	1729
Gender_female	4794
Gender_male	8637
Illumination_day	8136
Illumination_night/dusk	5295
Crash type (collision type)_head-on	1016
Crash type (collision type)_with pedestrian	3650
Crash type (collision type)_rear-end	810
Crash type (collision type)_run-off-road/fixed object/other	2423
Crash type (collision type)_side	4837
Crash type (collision type)_sideswipe	695
Weather_adverse	1038
Weather_clear	12,393
Area type_urban	10,492
Area type_non-urban	2939
Intersection_yes	5177
Intersection_no	8254
Vehicle type involved_bus/truck	643
Vehicle type involved_car	7111
Vehicle type involved_powered-two-wheelers (PTW)/bicycle	5478
Vehicle type involved_other/unknown	199

Table 2 – Temporal evolution of crash injury severity by year.

Year	Severity	
	SI	KSI
2006	1418	244
2007	1317	247
2008	1252	197
2009	1338	214
2010	1295	200
2011	1191	175
2012	1094	138
2013	1036	147
2014	936	78
2015	825	89

Overall, the mixed logit model showed good fit and was superior to the fixed effects model. More specifically, the log-likelihood ratio test of the null model versus the final model was found to be statistically significant at a 95% level (p -value < 0.001), as the log-likelihood value of the null model was -5157.1 and the respective value of the mixed logit model was -1621 . On the other hand, the log-likelihood value of the fixed effects binary logistic model was -4727 . Hence, the likelihood ratio between the fixed effects and the mixed logit model is significant at a 95% level, indicating that the mixed logit model outperforms the fixed effects model. An additional comparison by using the AIC shows similar results (the lower AIC indicates a better fit for the model). The AIC of the mixed logit model was found to be 3271.21, whilst the AIC of the fixed effects model was 9503.01. As a result, the mixed logit model is preferred over the fixed effects model.

Table 3 – Results of the mixed logit model (holistic model) for the period 2006–2015.

Variable	Beta coefficient	Standard error	p-value	Significance (%)
Constant	−1.110	0.149	<0.001	99
Standard deviation of constant	0.100	0.027	<0.001	99
Gender_female (reference category)	—	—	—	—
Gender_male	0.269	0.060	<0.001	99
Crash type_head-on (reference category)	—	—	—	—
Crash type_with pedestrian	−0.338	0.105	<0.001	99
Crash type_rear-end	−0.685	0.146	<0.001	99
Crash type_run-off-road/fixed object	0.311	0.096	<0.001	99
Crash type_side	−0.485	0.100	<0.001	99
Crash type_sideswipe	−0.690	0.158	<0.001	99
Time_day (reference category)	—	—	—	—
Time_night/dusk	0.298	0.054	<0.001	99
Location_non-urban (reference category)	—	—	—	—
Location_urban	−0.764	0.065	<0.001	99
Intersection_no (reference category)	—	—	—	—
Intersection_yes	−0.534	0.068	<0.001	99
Vehicle type_bus/truck (reference category)	—	—	—	—
Vehicle type_other/unknown	0.223	0.231	0.967	Non-significant
Vehicle type_car	−0.439	0.122	<0.001	99
Vehicle type_PTW/bicycle	0.293	0.124	0.019	95
Year_before 2010 (reference category)	—	—	—	—
Year_after 2010	−0.219	0.055	<0.001	99

A first inspection on the specific model output confirmed the hypothesis that there is unobserved heterogeneity across Greece counties, because the standard deviation of the constant term, which was allowed to vary across Greece counties, was found to be statistically significant at a 95% level (beta coefficient = 0.100, p -value < 0.001). However, it is noted that in model for the period 2011–2015, the standard deviation of constant term was only significant for 90% (p -value = 0.099). Nevertheless, unobserved heterogeneity is still confirmed.

The temporal evolution of crash injury severity outcomes is also investigated and confirmed, because the variable “year” is statistically significant (beta coefficient = −0.219, p -

value < 0.001), indicating a downward trend and less severe injury outcomes after 2010.

Furthermore, the temporal instability of crash-related factors in injury severities is confirmed by past relevant literature in the field (Behnood and Mannering, 2015, 2016). In our study, the log-likelihood of the holistic model was −1621, while the log-likelihood of the models before 2010 and after 2010 were −1420 and −1137, respectively. Therefore, the likelihood ratio test for the temporal instability was significant at a 95% level, confirming that it is more appropriate to apply separate models for before and after 2010, rather than having a holistic model for the whole

Table 4 – Results of the mixed logit model for the period 2006–2010.

Variable	Beta coefficient	Standard error	p-value	Significance (%)
Constant	−1.023	0.174	<0.001	99
Standard deviation of constant	0.344	0.032	<0.001	99
Gender_female (reference category)	—	—	—	—
Gender_male	0.361	0.071	<0.001	99
Crash type_head-on (reference category)	—	—	—	—
Crash type_with pedestrian	−0.376	0.122	0.002	99
Crash type_rear-end	−0.518	0.161	0.001	99
Crash type_run-off-road/fixed object	0.417	0.111	<0.001	99
Crash type_side	−0.411	0.112	<0.001	99
Crash type_sideswipe	−0.535	0.169	0.002	99
Time_day (reference category)	—	—	—	—
Time_night/dusk	0.345	0.062	<0.001	99
Location_non-urban (reference category)	—	—	—	—
Location_urban	−0.669	0.075	<0.001	99
Intersection_no (reference category)	—	—	—	—
Intersection_yes	−0.531	0.077	<0.001	99
Vehicle type_bus/truck (reference category)	—	—	—	—
Vehicle type_other/unknown	0.412	0.268	0.124	Non-significant
Vehicle type_car	−0.396	0.144	0.006	99
Vehicle type_PTW/bicycle	0.336	0.148	0.023	95

Table 5 – Results of the mixed logit model for the period 2011–2015.

Variable	Beta coefficient	Standard error	p-value	Significance (%)
Constant	–1.312	0.249	<0.001	99
Standard deviation of constant	0.100	0.061	0.099	90
Gender_female (reference category)	–	–	–	–
Gender_male	0.296	0.096	0.002	99
Crash type_head-on (reference category)	–	–	–	–
Crash type_with pedestrian	–0.297	0.176	0.091	90
Crash type_rear-end	–1.014	0.266	<0.001	99
Crash type_run-off-road/fixed object	0.218	0.165	0.188	Non-significant
Crash type_side	–0.551	0.174	0.001	99
Crash type_sideswipe	–1.034	0.293	<0.001	99
Time_day (reference category)	–	–	–	–
Time_night/dusk	0.330	0.089	<0.001	99
Location_non-urban (reference category)	–	–	–	–
Location_urban	–0.780	0.110	<0.001	99
Intersection_no (reference category)	–	–	–	–
Intersection_yes	–0.475	0.110	<0.001	99
Vehicle type_bus/truck (reference category)	–	–	–	–
Vehicle type_other/unknown	–0.101	0.404	0.804	Non-significant
Vehicle type_car	–0.479	0.198	0.016	95
Vehicle type_PTW/bicycle	0.356	0.202	0.078	90

period 2006–2015. Further comparisons were performed by using BIC test, confirming the temporal instability hypothesis. The BIC value of the holistic model was 3383.79, while the BIC of the model before 2010 and after 2010 were 2965.23 and 2394.67, respectively. It is noted that the same variables were used across all three separate models and mostly similar results were observed.

Regarding the individual person characteristics, it was found that males are usually associated with more severe injuries than females (p -value < 0.001). This finding can be considered in line with a few past studies in the field, such as [Koopmans et al. \(2015\)](#).

Regarding crash type, as expected, rear-end, side and sideswipe collisions result in lower injury severities. An interesting remark of our study is that crashes involving pedestrians are associated with lower child/adolescent severity than head-on collisions and run-off-road collisions/collisions with fixed objects. The increased severity outcomes of head-on and run-off-road collisions could be attributed to the potential high speeds and forces during crash, despite the improvements in car design standards over the past years and to the presence of heavy obstacles (trees, poles) near the roads. However, an interesting finding regards the statistically insignificant effect of run-off-road collisions/collisions with fixed objects, when different time periods are considered. Although the overall model (2006–2015), as well as the model regarding 2006–2010, indicated a consistently significant effect at a 95% level, a non-significant effect (p -value = 0.188) was found in the model utilizing 2011–2015 data, thus justifying the temporal instability of this risk factor. To further understand the impact of this particular risk factor, further investigation is needed in future studies in the field.

As for the crash environment characteristics, interesting trends are generally observed. Firstly, the positive sign of the beta coefficient of time variable (p -value < 0.001) shows that night crashes are usually more severe than day crashes, possibly due to reduced lighting conditions and illumination.

Furthermore, because of lower traffic flows during the night, drivers are generally more likely to speed and thus increasing the risk of more severe injuries if a crash occurs.

On the other hand, adverse weather conditions were not found to have any influence on injury severity of children and adolescents. This finding is also observed in a few similar past studies from Greece ([Theofilatos, 2017](#)). One possible explanation is that adverse weather conditions are not very frequently occurring in Greece. As a consequence, drivers might be surprised and thus compensate for these adverse conditions by driving more carefully (especially when children and adolescents are present in the car).

Crashes occurring at intersections have a negative beta coefficient and are associated with slighter injury outcomes than crashes outside intersections, perhaps due to the lower speeds at intersections. Interestingly, crashes that occur in urban areas tend to be less severe than crashes outside urban areas. This finding could be attributed to improper lighting conditions or poor road maintenance in rural areas and increased driving speeds in highways and rural areas. Generally, crashes on roads with higher speed limits tend to result in more severe injuries, probably because of the greater expected impact speeds. However, this result should be interpreted with care, because due to data limitations, it was not possible to distinguish between rural roads and highways. Definitely, this issue needs further investigation.

Lastly, the type of vehicle was found to have a strong influence on injury severity. Regarding car occupants, because bus/truck category is the reference category and the beta coefficient of car category was negative, children and adolescents are more likely to be slightly injured in bus crashes compared to car crashes. As expected, children and adolescents who are occupants in two-wheeler crashes (bicycles, mopeds and motorcycles) were more likely to sustain severe and fatal injuries than those involved in car and bus crashes. However, although this variable was strongly significant at a 95% level both in the overall model and the model for

2006–2010, when the period 2011–2015 was considered it was only significant at a 90% level (p -value = 0.078). Nevertheless, it is obvious that the lack of protection in powered-two-wheelers (PTWs) makes PTW occupants more vulnerable compared to car and heavy vehicle occupants. These findings also emphasize the need to raise awareness towards young people especially when they ride or are passengers in two-wheelers.

Overall, the analysis of injury severity of children and adolescents unveiled the potential risk factors and provided insights on their temporal instability. A final note on the results is that although not explicitly obvious in the current study results due to data limitations, there is also a strong need to launch campaigns in order to ensure that child restraint systems are properly used, since such measures have been found to have positive effect on child restraint used (Aigner-Breuss and Pilgerstorfer, 2017).

4. Conclusions

Despite that a large volume of studies in the past have analyzed the factors contributing to injury outcomes of crashes, relatively little research has been carried out regarding the injury severity analysis of children and adolescents. Our study uses a large national sample of disaggregate road crash data from all the regions in Greece for the period 2006–2015, including, a total of 13,431 reported injuries. In order to account for unobserved heterogeneity, a mixed logit model where the constant term followed a normal distribution is applied in which the year was also included as an independent variable. To the best of knowledge, this is the first study to have a special focus on factors affecting child and adolescent injury severity in traffic crashes by simultaneously addressing potential unobserved heterogeneity and temporal instability of crash injury related risk factors.

The estimation results showed that night crashes, crashes outside urban areas, run-off-road collisions/collisions with fixed objects as well as crashes involving males, bicycles or PTWs are generally associated with higher injury severity of children and adolescents. Interestingly, crashes involving pedestrians are associated with lower child/adolescent severity than head-on collisions and run-off-road collisions and collisions with fixed objects. The temporal instability of risk factors is confirmed through the application of likelihood ratio tests. For example, the run-off-road/fixed objects collisions variable showed strongly instable and inconsistent effect through different time periods.

The study shows a need to focus on policy measures, such as law enforcement especially when adolescents are riding a bicycle/PTW or when children and adolescents are involved in crashes during night trips and outside urban areas. Awareness raising and campaigns as well as intervention programs to increase safe teen driving outside urban areas are also needed to address potential risk factors associated with rural and freeway environments. Additionally, there is a need to launch campaigns in order to ensure that child restraint is properly used.

It was further demonstrated that injury severity is systematically reduced after 2010 in Greece. Indeed, despite the

economic crisis after 2010 in Greece, there was a huge effort by policy makers and road safety experts, mainly through the development of hundreds of kilometers of new motorways, improved and systematic road maintenance, new laws and enforcement. Of course, development in crashworthiness of new vehicles should be also considered to have a positive influence.

Future studies could utilize more recent data concerning the time period after the financial crisis or during the Corona Virus Disease 2019 (COVID-19) pandemic and compare with the current study findings. In that context, temporal instability could be further explored by considering more time periods or by applying additional likelihood ratio tests.

Although the utilized dataset was rich, more in-depth data that would enhance understanding the maneuvers, kinematics and driver behavior of crashes involving young people are needed as well. Due to data limitations, the present study examined two severity outcomes; however, data from other countries with more categories could be explored and provide additional evidence on children and adolescent injuries. It would also be interesting to develop similar models for adults involved in road crashes in Greece and compare the results with the present paper's findings. More in-depth analysis considering age, experience or gender differences should also be utilized.

Summing up, the findings of this study can assist in unveiling crash risk factors and prioritize programs and measures to promote road safety of children and adolescents in Greece, so as to tackle this problem even though there is a decreasing trend in the total number of injuries each year in Greece.

Conflict of interest

The authors do not have any conflict of interest with other entities or researchers.

Acknowledgments

This project has received funding from the European Union's Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 754462.

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